1. High-Correlation Feature Filter

Initialization

- Correlation magnitude upper limit $\tau_{corr} \in [0, 1]$.

Fitting Process

Inputs: Data matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$.

Procedure:

1. Compute $D \times D$ Pearson correlation matrix $\mathbf{C}$:

$$C_{ij} = \frac{\sum_{k=1}^{N}(X_{ki} - \bar{X}_i)(X_{kj} - \bar{X}_j)}{\sqrt{\sum_{k=1}^{N}(X_{ki} - \bar{X}_i)^2}\sqrt{\sum_{k=1}^{N}(X_{kj} - \bar{X}_j)^2}}$$

2. Identify set of feature indices for removal $\mathcal{I}_{drop}$:

$$\mathcal{I}_{drop} = \{j \mid \exists i < j \text{ such that } |C_{ij}| > \tau_{corr}\}$$

Transformation Process

Inputs: Data matrix $\mathbf{X}' \in \mathbb{R}^{N' \times D'}$, set $\mathcal{I}_{drop}$.

Procedure: Remove columns from $\mathbf{X}'$ with indices in $\mathcal{I}_{drop}$ (if valid).

2. Sequential Preprocessing Transformer

A pipeline applying transformations sequentially: $\mathbf{X} \xrightarrow{T_1} \mathbf{X}^{(1)} \xrightarrow{T_2} \mathbf{X}^{(2)} \xrightarrow{T_3} \mathbf{X}^{(3)} \xrightarrow{T_4} \mathbf{X}^{(4)}$.

Stage 1: Low-Variance Feature Removal (T_1)

- For each feature $\boldsymbol{x}_j$ in input $\mathbf{X}^{(0)}$, calculate sample variance $s_j^2 = \frac{1}{N-1}\sum_{k=1}^{N}(X_{kj}^{(0)} - \bar{X}_j^{(0)})^2$.
- Remove feature $\boldsymbol{x}_j$ if $s_j^2 < \tau_{var}$ (e.g., $\tau_{var} = 0.05$).
- Output: $\mathbf{X}^{(1)}$.

Stage 2: High-Correlation Feature Filtering (T_2)

- Applies the filter from Section 1 to $\mathbf{X}^{(1)}$ (e.g., with $\tau_{corr} = 0.95$).
- Output: $\mathbf{X}^{(2)}$.

Stage 3: Data Scaling (Standardization) (T_3)

- For each feature $\boldsymbol{x}_j^{(2)}$ in $\mathbf{X}^{(2)}$ with mean $\bar{X}_j^{(2)}$ and std. dev. $s_j^{(2)}$:

$$X_{kj}^{(3)} = \frac{X_{kj}^{(2)} - \bar{X}_j^{(2)}}{s_j^{(2)}}$$

  (If $s_j^{(2)} = 0$, $X_{kj}^{(3)} = 0$).

- Output: $\mathbf{X}^{(3)}$.

Stage 4: Principal Component Analysis (PCA) (T_4)

- Input $\mathbf{X}^{(3)}$ (assumed centered or centered internally).
- Compute covariance matrix $\mathbf{S}_X = \frac{1}{N-1}(\mathbf{X}^{(3)})^T\mathbf{X}^{(3)}$.
- Perform eigen-decomposition of $\mathbf{S}_X$ to get eigenvalues λ_i and eigenvectors $\boldsymbol{w}_i$.
- Select D_{pca} eigenvectors $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_{D_{pca}}$ corresponding to largest eigenvalues. Form projection matrix $\mathbf{W} = [\boldsymbol{w}_1, \ldots, \boldsymbol{w}_{D_{pca}}]$.
- Transformed data: $\mathbf{X}^{(4)} = \mathbf{X}^{(3)}\mathbf{W}$.
- Output: $\mathbf{X}^{(4)}$.